CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

May 2012

ME102 Fundamentals of Mechanical Engineering

Date: 18/5/12 **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain what is prime movers and also explain Enthalpy, Pressure and Force [04]

(b) Explain boyles law and charles law [03]

Q - 2 (a) Derive Characteristics Equation for a Perfect Gas [04]

(b) Explain working of centrifugal compressor. [05]

(c) Give the classification of I.C engines. [05]

OR

Q - 2 (a) Derive and expression for air standard efficiency for diesel cycle [04]

(b) What is priming ? and also Explain working of gear pump with sketch [05]

(c) Explain difference between four stroke and two stroke cycle and also fro clutch and brake. [05]

Q - 3 (a) Explain flexible coupling with neat sketch. [04]

or

(a) Explain elements of power transmission. [04]

(b) a six cylinder gasoline engine operates the four stokes cycle .the bore of each cylinder is 80mm and stroke 100mm the clearance volume per cylinder is 70cc at speed of 4000 rpm the fuel consumption is 20kg/hour and torque developed 150 Nm. [06]

Calculate (1)brake power(2) brake mean effective pressure(3) brake thermal efficiency if the calorific value of fuel is 43000 kj/kg fin relative efficiency on brake power basis assuming the engine work on constant cycle ,γ =1.4 for air.

(c) Explain sources of fuel and what is CNG? [04]

SECTION – II

Q - 4 (a) Define the following terms (i) Dryness fraction (ii) Enthalpy of evaporation (iii) Degree of Superheat. [03]

(b) Explain with neat sketch (i) Oldham coupling (ii) Band brake [04]

Q - 5 (a) What is the difference between the boiler accessories and boiler mountings? Explain in brief with neat sketch Air Preheater. [04]

(b) What is function of clutch in an automobile? List different types of clutches used in automobiles. [03]

(c) A steam boiler evaporates 17000 kg/hr of steam at 12 bar and quality of 0.97 from feed water at 55°C. When the coal is fired at a rate of 2040kg/hr having calorific value 27400 kJ/kg. Find (i) Heat supplied per hour (ii) Boiler efficiency (iii) Equivalent evaporation. [07]

OR

Q - 5 (a) Explain with neat sketch the working and construction of Lancashire boiler. [04]

(b) What is function of coupling? Explain any one type of coupling used to connect two shafts. [03]

(c) Following data is given for throttling calorimeter : [07]
Pressure before throttling = 14 bar
Pressure after throttling = 01 bar
Temperature after throttling = 120°C ; Cps = 2.1 kJ/kg
Calculate (i) Dryness fraction of steam before passing through the calorimeter.
(ii) Minimum Value of dryness fraction when the pressure before and after throttling are same.

Q - 6 (a) Explain with neat sketch working and construction of barrel calorimeter. [04]

(b) Classify the belt drive. [03]

(c) Explain in brief the working and construction of window air conditioner. Define C.O.P and Unit of Refrigeration. [07]

OR

Q - 6 (a) Explain the working of separating calorimeter in brief with neat sketch. [04]

(b) What do you understand by gear train? [03]

(c) Explain vapour compression refrigeration cycle with neat sketch and state function of individual components of vapour compression refrigeration system. [07]

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